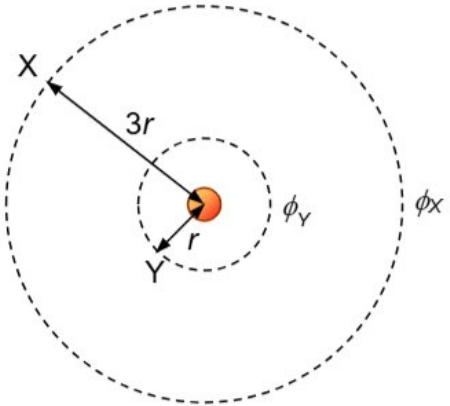
10 Two points X and Y are on equipotential lines with potentials ϕ_X and ϕ_Y which are at distances $3r$ and r respectively from a star.



Which statement is correct?

A The ratio of $\phi_X : \phi_Y$ is 3 : 1.

B The ratio of $\phi_X : \phi_Y$ is 9 : 1.

C Work done by gravitational field to move a mass from X to Y is positive.

D Work done by external agent to move a mass from Y to X is negative.